Sympathomimetic drugs

Hawler Medical University
College of Medicine
Department of Pharmacology

Dr. Susan Abdulkadir Farhadi
MSc, PhD Pharmacology

The sympathetic nervous system is an important regulator of virtually all organ systems. This is particularly evident in the regulation of blood pressure.

The ultimate effects of sympathetic stimulation are mediated by release of norepinephrine from nerve terminals, which then activates adrenoceptors on postsynaptic sites.

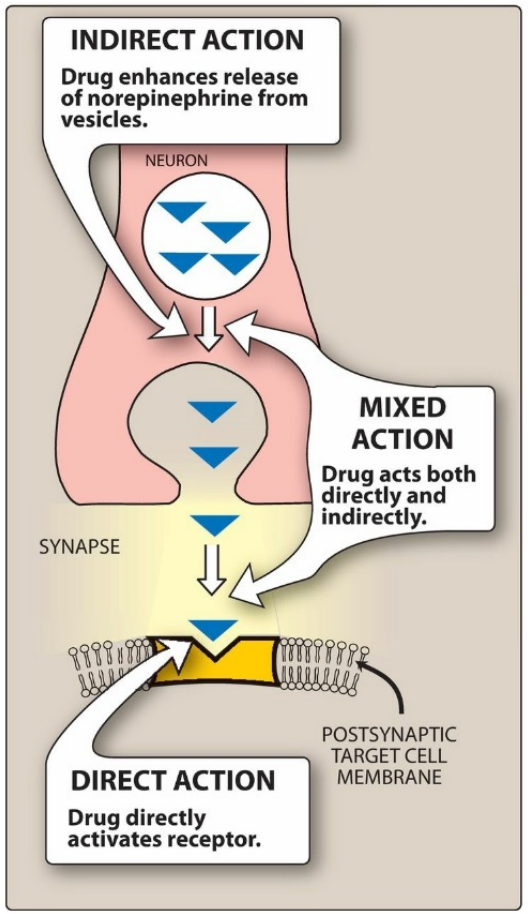
Also, in response to a variety of stimuli such as stress, the adrenal medulla releases epinephrine, which is transported in the blood to target tissues. In other words, epinephrine acts as a hormone, whereas norepinephrine acts as a neurotransmitter.

Mechanism of action of adrenergic agonists

1- Directly acting

2- Indirectly acting

3- Mixed action



1. Direct-acting agonists

- These drugs act directly on α or β receptors, producing effects similar to those that occur following stimulation of sympathetic nerves or release of epinephrine from the adrenal medulla.
- Examples of direct-acting agonists include epinephrine, norepinephrine, isoproterenol, dopamine, and phenylephrine.

2. Indirect-acting agonists

- These agents may block the reuptake of norepinephrine or cause the release of norepinephrine from the cytoplasmic pools or vesicles of the adrenergic neuron.
- The norepinephrine then traverses the synapse and binds to α or β receptors.
- Examples of reuptake inhibitors and agents that cause norepinephrine release include cocaine and amphetamine, respectively.

3. Mixed-action agonists

- Ephedrine and its stereoisomer, pseudoephedrine, both stimulate adrenoceptors directly and enhance release of norepinephrine from the adrenergic neuron

1) Directly acting Sympathomimetic drugs

A) Catecholamines:

Adrenaline, Noradrenaline and Dopamine (natural)

Isoprenaline and Dobutamine (synthetic)

- Adrenaline: α1, α2, β1, β2 and β3 agonist
- Noradrenaline: α1, α2, β1 and β3 agonist
- Isoprenaline: Pure β agonist
- Dopamine: α1, β1 and it's a Dopamine receptor agonist (like fenoldopam)
- Dobutamine: β1 receptor agonist

Pharmacological effects of Catecholamines
Effects on CVS:

- Heart
- Blood vessels
- Blood pressure (BP)

A-The heart has only one adrenergic receptor type, **β1**, which is excitatory. It causes an increase in heart rate (tachycardia; **chronotropic** effect) and cardiac contractility (**inotropic** effect). These two factors contribute to **increasing cardiac output**.

- Catecholamines that stimulate the β1 receptor are (all 5):
  - Adrenaline
  - Noradrenaline
  - Dopamine
  - Dobutamine
  - Isoprenaline

B- Blood vessels of the skin and mucous membranes only have one type of adrenergic receptors, α1, which are excitatory. Stimulation of the α1 receptors in these vessels causes **vasoconstriction** and hence **decreased blood supply**. These receptors are stimulated by:

- Adrenaline
- Noradrenaline
- Dopamine

- **Visceral blood vessel** have α1 and β2 receptors.
  - Isoprenaline (stimulates β2) causes vasodilation
  - Noradrenaline (stimulates α1) causes vasoconstriction
  - Adrenaline (stimulates both β2 and α1) causes vasoconstriction if at a high dose, and vasodilation at a relatively low dose.
  - Dopamine (stimulates D1 and α1) causes vasoconstriction at high doses, and vasodilation in the kidney at a relatively low dose.

- *β2 activation threshold < α1 activation threshold*
- *At high doses: α1 stimulation dominates and vasoconstriction results.*
- *At low doses: β2 stimulation dominates and vasodilation results.*

- At high doses of dopamine, the α1 stimulation is stronger than the D1 stimulation, so vasoconstriction results. Selective dopamine receptor agonists, which only stimulate D1, e.g., fenoldopam, cause vasodilation only. Fenoldopam is used as an antihypertensive agent.

C- Effects on blood pressure (BP):

- **Systolic blood pressure** is mainly affected by the **cardiac output**. They're **directly related;** If the cardiac output increases, the systolic blood pressure increases
- **Diastolic blood pressure** is mainly affected by **peripheral vascular resistance.** They're **directly related;** if the peripheral vascular resistance increases, the diastolic blood pressure increases.
  - Vasodilation → peripheral vascular resistance decreases → diastolic blood pressure decreases.
  - Vasoconstriction → peripheral vascular resistance increases → diastolic blood pressure increases.

The effects of the adrenergic receptors on blood pressure:

Isoprenaline:

- Cardiac output increases (β1) → systolic BP increases
- Causes vasodilation (β2) → peripheral vascular resistance decreases → diastolic BP decreases

Adrenaline:

- Increases cardiac output (β1) → systolic BP increases
- At high doses, causes vasoconstriction (α1 > β2) → peripheral vascular resistance increases → diastolic BP increases
- At relatively low doses, causes vasodilation (β2) → peripheral vascular resistance decreases → diastolic BP decreases

- Noradrenalin:
  - Increases cardiac output (β1) → systolic BP increases
  - Causes vasoconstriction (α1) → peripheral vascular resistance increases → diastolic BP increases
    - Mean blood pressure increases

- Noradrenaline increases the mean blood pressure of the CVS, and this stimulates baroreceptors (mechanoreceptors that are sensitive to changes in pressure), causing a negative feedback loop that causes a decrease in heart rate (bradycardia). Although noradrenaline has chronotropic effects by itself, the inhibitory feedback signal is stronger than the β1 stimulation, so bradycardia occurs instead.

The effects of catecholamines on non-vascular smooth muscles:

1- Effects on urinary bladder smooth muscles

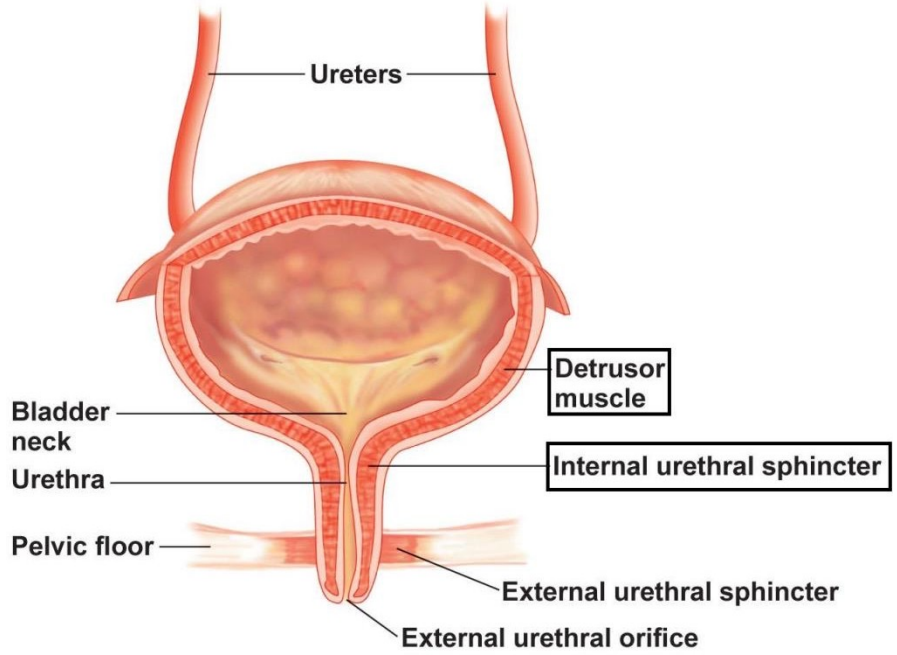
The urethra has detrusor muscles and the internal urethral sphincter, both of which are made out of smooth muscle tissue.

- The detrusor muscle has

β2 and β3 receptors, which are inhibitory and cause it to relax

- The internal urethral sphincter has α1 receptors, which are stimulatory and cause it to contract

Adrenaline and noradrenaline, for example, cause detrusor relaxation and the constriction of the internal urethral sphincter, leading to urinary retention.



2- GI smooth muscles: these smooth muscles have both α1 and β2 receptors, both of which are inhibitory and cause relaxation.

3- Uterine smooth muscles: have β2 receptors, Adrenaline produces relaxation in pregnant women.

4- Bronchial muscles: have β2 receptors, which are inhibitory and produce bronchodilation.

The metabolic effects of catecholamines:

- Adrenaline has a significant hyperglycemic effect because of increased glycogenolysis in the liver (β2 effect), increased release of glucagon (β2 effect), and a decreased release of insulin (α2 effect).
- Adrenaline causes lipolysis of triglycerides (via β3 receptors). β blockers cause increased levels of triglycerides, so they are contra-indicated in hyperlipidemia.
- Stimulation of β 1 receptor in the kidney results in the release of Renin.

Other effects of catecholamines:

The eyes: stimulation of α1 receptors in the radial dilatory muscles causes them to contract, thereby causing mydriasis (pupil dilation).

Note: large doses of adrenaline can cause exophthalmos (forward protrusion of the eye) by stimulating the orbital muscles.

Blood: adrenaline causes an increase in factor V, thereby facilitating coagulation.

Skeletal muscles: Adrenaline promotes neuromuscular transmission at neuromuscular junctions (i.e., it increases ACh release at these junctions)

Electrolytes: Chronic β2 stimulation may cause hypokalemia.

Salivary glands: sympathetic stimulation of salivary glands causes them to produce a thick, viscous secretion (as opposed to the copious, watery secretion they produce in response to parasympathetic stimulation).

Therapeutic uses of catecholamines:

- Catecholamines (e.g., adrenaline, noradrenaline and dopamine) are used for the treatment of shock. Adrenaline increases systolic pressure through stimulating of B1 receptor and increases diastolic blood pressure through stimulating of alpha 1 receptor.
- Dopamine is the drug of choice for shock (cardiogenic and septic). It raises blood pressure by increasing cardiac output (Stimulation of B1 receptor) and peripheral vascular resistance by causing vasoconstriction (Through stimulation of alpha 1 receptor. In addition, it also causes vasodilation of renal blood vessels (Stimulation of D1 receptor); therefore this drug unlike the other catecholamines it can increase renal blood flow.

- Dopamine and Dobutamine are given parenterally to treat acute heart failure.
- Adrenaline is used to treat cardiac arrest, and is given in this case via an injection to the heart.
- Adrenaline prolongs the effects of local anaesthetics by causing vasoconstriction and thereby making the anaesthetic persists longer at the injection site before being absorbed into systemic circulation (this also helps decrease the side effects of the local anaesthetic produced at other sites)
- Adrenaline (locally) can be used to decrease bleeding as nasal bleeding by producing vasoconstriction

The pharmacokinetics of catecholamines:

- Catecholamines are given parenterally rather than orally, as they are not absorbed well when given orally. They have a short duration of action and are rapidly degraded by mono-amino oxidase (MAO) and Catechol-O-methyltransferase (COMT).

Adverse effects of catecholamines:

- Hypertension
- Tachycardia
- Cerebral bleeding/hemorrhage (due to increased blood pressure)
- Hyperglycemia (adrenaline)

Non-catecholamines:

α-receptor agonists:

- Phenylephrine
- Methoxamine
- Oxymetazoline
- Xylometazoline
- Naphazoline

Clinical use:

- These can cause vasoconstriction in the nose, so they are used as nasal decongestants. They are given orally or applied topically (e.g., as nasal sprays or drops).
- Phenylephrine is α1-selective. It can be used to treat hypotension in hospitalized or surgical patients (especially those with tachycardia, as phenylephrine doesn't stimulate β1 and therefore has no direct effect on the heart. It does, however, induce bradycardia as a response to the increased blood pressure it produces.) It's also used as a nasal decongestant (given orally or topically) as well as for mydriasis (in ophthalmic solutions).
- **Side effects:** they increase blood pressure and are contraindicated in hypertension.

β1-receptor agonists:

- Dobutamine is a synthetic catecholamine that is related to dopamine. It is administered by i.v. infusion because of its short duration (2 min).
- It activates mainly cardiac β1-receptors with no effect on dopamine receptors leading to increase cardiac output with little or no vascular effects.
- It is given by continuous i.v. infusion in cardiogenic shock (a complication of left ventricular infarction), to reverse myocardial depression and increase cardiac output COP .

β2-receptor agonists:

- Salbutamol (albuterol)
- Terbutaline
- Salmeterol

- In pregnant women, since these cause relaxation of uterine muscles, they cause a delay in delivery (they can be used to delay premature labor).
- **Clinical use:** these β2-selective receptors cause bronchodilation and are the drug of choice for asthma treatment.

β3-receptor agonist: mirabegron.

- **Clinical use:** Mirabegron relaxes bladder muscles and is used to increase the capacity of the urinary bladder for patients with overactive bladders.

Indirectly acting sympathomimetic agents:

Amphetamine (increases the release of noradrenaline):

- Increases blood pressure (so it's contraindicated in hypertension)
- Increases cardiac output
- It also has central effects as an (indirect) CNS stimulant, increasing mental and physical activity.
  - It decreases fatigue (anti-fatigue agent)
  - It decreases appetite (anorexiant)
  - It decreases sleepiness (causes insomnia)

  Clinical uses of amphetamine:
- **Appetite suppression:** fenfluramine and phentermine are derivatives of amphetamine.
- Used for neurological diseases, e.g., narcolepsy (sleep attacks) and ADHD.

Adverse effects of amphetamine:

- Addiction and tolerance (requiring larger and larger doses for the same, previous effects as time passes on)
- Increased cardiac output
- Increased blood pressure, so amphetamine is contraindicated in patients with hypertension and/or cardiovascular disease.
- Anorexia: Fenfluramine and phentermine

Mixed-acting adrenergic agonists:

- Ephedrine and pseudoephedrine
- Pseudoephedrine is used as a nasal decongestant. It’s similar to ephedrine but with less CNS effects. Ephedrine can be used for hypotension
- Ephedrine is a potential drug of abuse.
- Pseudoephedrine can also be used to produce methamphetamine, a powerful drug of abuse.